

Brownian Motion and Markov Property

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1 Brownian Motion

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let

$$I \subset \mathbb{R}_+ := [0, \infty)$$

be an index set. A stochastic process is a family of random variables

$$(X_t)_{t \in I}$$

defined on $(\Omega, \mathcal{F}, \mathbb{P})$. For each time $t \in I$,

$$X_t : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

is measurable. For $\omega \in \Omega$, the map

$$t \mapsto X_t(\omega)$$

is called a path.

Definition 1 (Brownian motion). *A process $(B_t)_{t \geq 0}$ is Brownian motion if:*

- (i) $\mathbb{P}(\{\omega \in \Omega : B_0(\omega) = 0\}) = 1$;
- (ii) for $0 \leq s < t$, $B_t - B_s \sim N(0, t - s)$;
- (iii) (B_t) has independent increments;
- (iv) almost all sample paths are continuous.

Proposition 1. *If (B_t) is Brownian motion, then:*

- (i) for fixed $t_0 \geq 0$, the process

$$\tilde{B}_t := B_{t+t_0} - B_{t_0}$$

is Brownian motion;

- (ii) for $\lambda > 0$, the process

$$\tilde{B}_t := \frac{1}{\sqrt{\lambda}} B_{\lambda t}$$

is Brownian motion.

Proof. The conditions (i), (iii), and (iv) are immediate. For (ii), in the first case the increment law is trivial. In the second case,

$$\frac{1}{\sqrt{\lambda}}(B_{\lambda t} - B_{\lambda s}) \sim N(0, t - s),$$

so condition (ii) holds. □

Let

$$\mathcal{T}_n = \{0 = t_0^{(n)} < t_1^{(n)} < \dots < t_{N(\mathcal{T}_n)}^{(n)} < \infty\}$$

be a sequence of partitions of $[0, \infty)$ such that

$$|\mathcal{T}_n| := \sup_{1 \leq i < N(\mathcal{T}_n)} |t_{i+1}^{(n)} - t_i^{(n)}| \rightarrow 0, \quad t_{N(\mathcal{T}_n)}^{(n)} \rightarrow \infty.$$

Theorem 1. *For Brownian motion (B_t) :*

(i) for every $t \geq 0$,

$$S_t^n := \sum_{\substack{t_i^{(n)} \in \mathcal{T}_n \\ t_i^{(n)} \leq t}} \left(B_{t_{i+1}^{(n)}} - B_{t_i^{(n)}} \right)^2 \longrightarrow t$$

in $L^2(\Omega, \mathcal{F}, \mathbb{P})$;

(ii) if $\mathcal{T}_n \subset \mathcal{T}_{n+1}$, or if $\sum_{n \geq 1} |\mathcal{T}_n| < \infty$, then

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} S_t^n = t, \forall t \geq 0 \right) = 1.$$

Very nontrivial fact that holds in Brownian motion!

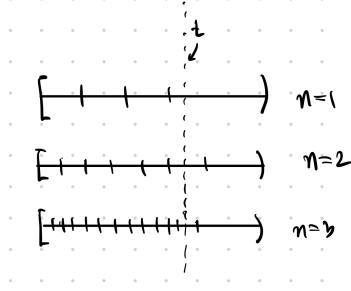


Figure 1: Visualization of Partition

Proof. For (i),

$$\begin{aligned} \mathbb{E}[S_t^n] &= \sum_{\substack{t_i^{(n)} \in \mathcal{T}_n \\ t_i^{(n)} \leq t}} \mathbb{E} \left[\left(B_{t_{i+1}^{(n)}} - B_{t_i^{(n)}} \right)^2 \right] \\ &= \sum_{\substack{t_i^{(n)} \in \mathcal{T}_n \\ t_i^{(n)} \leq t}} \left(t_{i+1}^{(n)} - t_i^{(n)} \right) \longrightarrow t. \end{aligned}$$

Moreover,

$$\begin{aligned} \mathbb{E} \left[(S_t^n - \mathbb{E}[S_t^n])^2 \right] &= \text{var}(S_t^n) \\ &= \sum_{\substack{t_i^{(n)} \in \mathcal{T}_n \\ t_{i+1}^{(n)} \leq t}} \text{var} \left(\left(B_{t_{i+1}^{(n)}} - B_{t_i^{(n)}} \right)^2 \right) \\ &= \sum_i 2 \left(t_{i+1}^{(n)} - t_i^{(n)} \right)^2 \longrightarrow 0, \end{aligned}$$

using independence of increments. Hence

$$\mathbb{E}[S_t^n] \rightarrow t, \quad S_t^n - \mathbb{E}[S_t^n] \rightarrow 0 \quad \text{in } L^2(\Omega, \mathcal{F}, \mathbb{P}).$$

For (ii), assume first that $\mathcal{T}_n$ is the dyadic partition

$$t_i^{(n)} = \frac{i-1}{2^n}, \quad i = 1, \dots, n2^n + 1.$$

If $t \leq k \in \mathbb{N}$, then

$$\text{var}(S_t^n) \leq 2 \sum_{i=1}^{2^n t} \left(\frac{1}{2^n} \right)^2 \leq \frac{2k}{2^n}.$$

Thus by Chebyshev's inequality,

$$\mathbb{P}(|S_t^n - \mathbb{E}[S_t^n]| > a) \leq \frac{1}{a^2} \text{var}(S_t^n),$$

and

$$\sum_{n=1}^{\infty} \mathbb{P}(|S_t^n - \mathbb{E}[S_t^n]| > a) \leq \sum_{n=1}^{\infty} \frac{2k}{a^2 2^n} < \infty.$$

Therefore

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} |S_t^n - \mathbb{E}[S_t^n]| = 0\right) = 1, \quad \mathbb{P}\left(\lim_{n \rightarrow \infty} S_t^n = t\right) = 1.$$

Let Ω_t be the event that $\lim_{n \rightarrow \infty} S_t^n = t$. Define

$$\Omega' := \bigcap_{t \in \mathbb{Q}_+} \Omega_t.$$

Then $\mathbb{P}(\Omega') = 1$. Hence, for $\omega \in \Omega'$ and $t \in \mathbb{Q}_+$,

$$S_t^n(\omega) \rightarrow t.$$

For arbitrary $t \in [0, \infty) \setminus \mathbb{Q}_+$, take dyadic rationals $t_k^- \downarrow t$ and $t_k^+ \uparrow t$. Since S_t^n is increasing in t ,

$$S_{t_k^-}^n(\omega) \leq S_t^n(\omega) \leq S_{t_k^+}^n(\omega).$$

Sending $n \rightarrow \infty$ gives

$$t_k^- \leq \liminf_n S_t^n(\omega) \leq \limsup_n S_t^n(\omega) \leq t_k^+,$$

and then $k \rightarrow \infty$ gives $t = \lim_n S_t^n(\omega)$. □

Remark 1. Let $\Omega'' \subset \Omega$ be the event that $t \mapsto B_t(\omega)$ is continuous. On $\Omega' \cap \Omega''$, the following hold:

$$t \mapsto B_t(\omega) \text{ is continuous,}$$

and

$$\langle B(\omega) \rangle_t := \lim_{n \rightarrow \infty} \sum_{\substack{t_i^{(n)} \in \mathcal{T}_n \\ t_i^{(n)} \leq t}} \left(B_{t_{i+1}^{(n)}}(\omega) - B_{t_i^{(n)}}(\omega) \right)^2 = t, \quad \forall t \geq 0.$$

We call such a path a typical path, and this limit is the quadratic variation of the typical path.

Proposition 2. A typical path of Brownian motion is continuous and has unbounded variation on each interval $[0, T]$, $T > 0$.

Proof. For a partition $\mathcal{T}_n$,

$$\sum_{\substack{t_i^{(n)} \in \mathcal{T}_n \\ t_i^{(n)} \leq t}} \left(B_{t_{i+1}^{(n)}}(\omega) - B_{t_i^{(n)}}(\omega) \right)^2 \leq \sup_i |B_{t_{i+1}^{(n)}}(\omega) - B_{t_i^{(n)}}(\omega)| \sum_i |B_{t_{i+1}^{(n)}}(\omega) - B_{t_i^{(n)}}(\omega)|.$$

For fixed ω , uniform continuity on $[0, T]$ gives

$$\sup_i |B_{t_{i+1}^{(n)}}(\omega) - B_{t_i^{(n)}}(\omega)| \rightarrow 0.$$

If the total variation were bounded, the quadratic variation would be 0, contradicting $\langle B \rangle_t = t$. □

1.1 Construction of Brownian Motion

Idea. A time-consistent family of distributions, or marginals, determines a measure.

Let

$$\mathbb{R}^I := \{\omega : I \rightarrow \mathbb{R}\} \quad \text{be the function space.}$$

Cylinder sets are sets of the form

$$E_I = \{\omega \in \mathbb{R}^I : (\omega(t_1), \dots, \omega(t_n)) \in A_1 \times \dots \times A_n\},$$

where $A_i \in \mathcal{B}(\mathbb{R})$ and $t_i \in I$. The σ -algebra $\sigma(E_I)$ is the smallest σ -algebra containing all such cylinder sets; equivalently, it is the smallest σ -algebra making the evaluation maps

$$f_{t_1, \dots, t_n} : \mathbb{R}^I \rightarrow \mathbb{R}^n, \quad \omega \mapsto (\omega(t_1), \dots, \omega(t_n))$$

measurable.

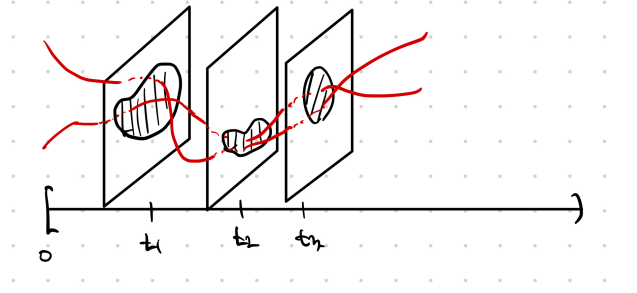


Figure 2: Measure constructed by consistent family

Theorem 2 (Kolmogorov extension theorem). *For a consistent family $\mu_{t_1, \dots, t_n}$ of probability measures on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$, there exists a probability measure $\mathbb{P}$ on $(\mathbb{R}^I, \sigma(E_I))$ such that*

$$\mu_{t_1, \dots, t_n} = \mathbb{P} \circ f_{t_1, \dots, t_n}^{-1}.$$

Let $I = \mathbb{R}_+ = [0, \infty)$. Define

$$p_t(x, y) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{1}{2} \frac{(y-x)^2}{t}\right), \quad t > 0,$$

and

$$p_0(x, y) dy = \delta_x(dy).$$

Then the measures

$$\mu_{t_1, \dots, t_n}(A_1 \times \dots \times A_n) = \int_{A_1} \dots \int_{A_n} \prod_{i=1}^n p_{t_i - t_{i-1}}(x_{i-1}, dx_i)$$

are consistent.

Therefore $\tilde{\mathbb{P}}$ on $(\mathbb{R}^{\mathbb{R}_+}, \sigma(E_{\mathbb{R}_+}))$ is defined by Kolmogorov's extension theorem. But we cannot immediately have property (iv) of Brownian motion, namely continuous paths.

So we detour to dyadic rationals. Restricting $t_1, \dots, t_n$ to $\mathbb{Q}_+$, we get a measure Q on

$$(\mathbb{R}^{\mathbb{Q}_+}, \sigma(E_{\mathbb{Q}_+}))$$

with marginal distributions $\mu_{t_1, \dots, t_n}$, $t_i \in \mathbb{Q}_+$.

Theorem 3. *Let*

$$C(\mathbb{Q}_+) := \bigcap_{n \geq 1} \left\{ \omega \in \mathbb{R}^{\mathbb{Q}_+} : \omega|_{[0, n] \cap \mathbb{Q}_+} \text{ is uniformly continuous} \right\}.$$

Then

$$C(\mathbb{Q}_+) \in \sigma(E_{\mathbb{Q}_+}), \quad Q(C(\mathbb{Q}_+)) = 1.$$

Good thing about dyadic rationals: since they are countable, almost all paths are continuous.

Assuming Theorem, we obtain Brownian motion. Let

$$\Omega = C(\mathbb{R}_+) = \{\omega : \mathbb{R}_+ \rightarrow \mathbb{R} \text{ continuous}\}.$$

Use the metric

$$\rho(\omega, \omega') = \sum_{n=1}^{\infty} 2^{-n} \left(1 \wedge \sup_{t \in [0, n]} |\omega(t) - \omega'(t)| \right).$$

Then Ω is complete and separable. Let

$$\mathcal{F} = \mathcal{B}(C(\mathbb{R}_+)).$$

The evaluation maps

$$B_t : \Omega \rightarrow \mathbb{R}, \quad \omega \mapsto \omega(t)$$

are measurable. Hence

$$\sigma(B_t : t \geq 0) \subset \mathcal{F}.$$

Since Ω is separable, let $D \subset \Omega$ be a countable dense subset. Then

$$\ell(\omega, \omega^0) = \sum_{n=1}^{\infty} 2^{-n} \left(1 \wedge \sup_{t \in [0, n] \cap \mathbb{Q}_+} |B_t(\omega) - \omega^0(t)| \right)$$

is $\sigma(B_t : t \in \mathbb{Q}_+)$ -measurable. Therefore the balls $B^\ell(\omega^0, r)$ are $\sigma(B_t : t \geq 0)$ -measurable, and hence

$$\mathcal{F} \subset \sigma(B_t : t \geq 0).$$

Consequently,

$$\mathcal{F} = \sigma(B_t : t \geq 0).$$

Define

$$\Psi : (C(\mathbb{Q}_+), C(\mathbb{Q}_+) \cap \sigma(E_{\mathbb{Q}_+}), Q) \rightarrow (\Omega, \mathcal{F})$$

by sending ω to its unique continuous extension $\bar{\omega}$. This map is measurable because inverse images of generators are measurable; for example,

$$\Psi^{-1}(\{B_t < a\}) = \bigcup_{k \in \mathbb{N}} \bigcup_{N \in \mathbb{N}} \bigcap_{m \geq n} \left\{ \omega \in C(\mathbb{Q}_+) : \omega(q_m) < a - \frac{1}{k} \right\},$$

where $q_m \downarrow t$ are dyadic rationals.

Set

$$\mathbb{P} = Q \circ \Psi^{-1}.$$

Claim. $B_t(\omega) = \omega(t)$, $t \geq 0$, $\omega \in \Omega = C(\mathbb{R}_+)$, is Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$.

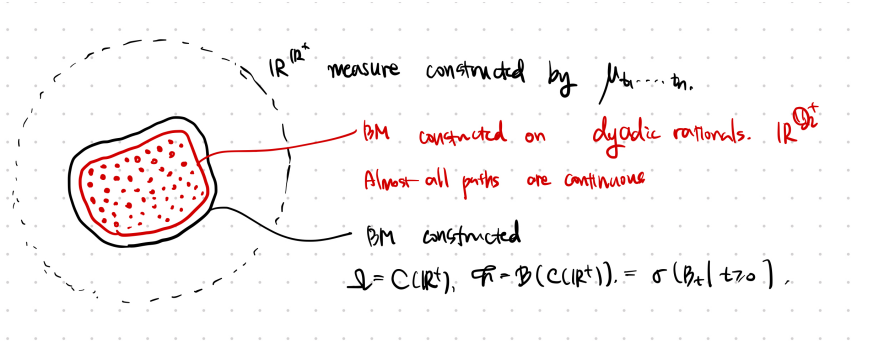


Figure 3: Proof Idea: Extension of dyadic numbers to real

We have

$$\Omega = C(\mathbb{R}_+), \quad \mathcal{F} = \mathcal{B}(C(\mathbb{R}_+)) = \sigma(B_t : t \geq 0), \quad \mathbb{P} = Q \circ \Psi^{-1}.$$

Check the finite-dimensional distributions:

$$\mathbb{P}((B_{t_1}, \dots, B_{t_n}) \in A_1 \times \dots \times A_n) = \int_{A_1} \dots \int_{A_n} \prod_{i=1}^n p_{t_i - t_{i-1}}(x_{i-1}, dx_i).$$

For $t_1, \dots, t_n \in \mathbb{Q}_+$ this holds by construction. For $t_1, \dots, t_n \in \mathbb{R}_+$ it also holds by approximation. In particular, for $n = 1$,

$$B_t \sim N(0, t), \quad B_0 = 0 \quad \mathbb{P}\text{-a.s.}$$

For $n = 2$, if

$$A_1 = \{(x, y) \in \mathbb{R}^2 : y - x \leq a\},$$

then

$$\mathbb{P}(B_{t_2} - B_{t_1} \leq a) = \int_{-\infty}^a \frac{1}{\sqrt{2\pi(t_2 - t_1)}} \exp\left(-\frac{z^2}{2(t_2 - t_1)}\right) dz.$$

Finally, independence of increments is obtained similarly.

Remaining proof of Theorem Let

$$\Omega_n := \left\{ \omega \in \mathbb{R}^{\mathbb{Q}_+} : \omega|_{[0,n] \cap \mathbb{Q}_+} \text{ is uniformly continuous} \right\}.$$

If $\Omega_n \in \sigma(E_{\mathbb{Q}_+})$ and $Q(\Omega_n) = 1$ for all n , then we are done. Notice that

$$\Omega_n = \bigcap_{M \geq 1} \bigcup_{N \geq 1} \bigcap_{\substack{t, s \in \mathbb{Q}_+ \cap [0, n] \\ |t-s| \leq 1/N}} \left\{ |X_t - X_s| \leq \frac{1}{M} \right\} \in \sigma(E_{\mathbb{Q}_+}).$$

Let

$$Y_t^n := \frac{1}{n} X_{nt}, \quad t \in \mathbb{Q}_+.$$

For $k \in \mathbb{N}$ and $0 \leq s < t \leq 1$ in $\mathbb{Q}_+$,

$$\begin{aligned} \mathbb{E}_Q[|Y_t^n - Y_s^n|^{2k}] &= \mathbb{E}_Q[|X_t - X_s|^{2k}] \\ &= \mathbb{E}_Q[|X_{t-s}|^{2k}] \\ &= (t-s)^k \mathbb{E}_Q[|X_1|^{2k}] = C_k |t-s|^k. \end{aligned}$$

By the Kolmogorov–Centsov theorem, there exists $\Omega^* \in \sigma(E_{\mathbb{Q}_+})$ with $Q(\Omega^*) = 1$ such that for $\omega \in \Omega^*$,

$$t \mapsto \frac{1}{\sqrt{n}} X_{nt}(\omega), \quad t \in [0, 1] \cap \mathbb{Q}_+,$$

is Hölder continuous of order $\delta \in (0, 1/2)$. Thus

$$t \mapsto X_t(\omega), \quad t \in [0, n] \cap \mathbb{Q}_+,$$

is Hölder continuous of order δ , and hence uniformly continuous. Therefore $\Omega^* \subset \Omega_n$ and $Q(\Omega_n) = 1$.

Theorem 4. Let $(X_t)_{t \in \mathbb{Q}_+}$ be a stochastic process on $(\mathbb{R}^{\mathbb{Q}_+}, \sigma(E_{\mathbb{Q}_+}), Q)$. Suppose

$$\mathbb{E}_Q[|X_t - X_s|^\alpha] \leq K |t-s|^{1+\beta}, \quad t, s \in \mathbb{Q}_+ \cap [0, 1].$$

If $\delta \in (0, \beta/\alpha)$, then there exists $\Omega^* \in \sigma(E_{\mathbb{Q}_+})$ with $Q(\Omega^*) = 1$ such that for $\omega \in \Omega^*$ there is $n^*(\omega) \in \mathbb{N}$ satisfying

$$\sup_{\substack{0 \leq t-s < 2^{-n^*(\omega)} \\ t, s \in \mathbb{Q}_+ \cap [0, 1]}} \frac{|X_t(\omega) - X_s(\omega)|}{|t-s|^\delta} \leq C := \frac{2}{1-2^{-\gamma}}.$$

In particular, every path is Hölder continuous of order δ on $\mathbb{Q}_+ \cap [0, 1]$.

1.2 The Markov Property

Let

$$\Omega = C(\mathbb{R}_+), \quad B_t(\omega) = \omega(t), \quad \mathcal{F} = \mathcal{B}(C(\mathbb{R}_+)) = \sigma(B_t : t \geq 0).$$

Let $\mathbb{P}_x$ be the measure starting at $x \in \mathbb{R}$. This assigns measure 0 to paths not starting at x . Write

$$\mathbb{E}_x[Y] = \int_{\Omega} Y(\omega) \mathbb{P}_x(d\omega).$$

Let

$$\mathcal{F}_t^0 := \sigma(B_u : u \leq t).$$

This is a filtration on $\mathcal{F}$. Define the shift

$$\theta_s : \Omega \rightarrow \Omega, \quad \theta_s(\omega)(t) = \omega(s+t), \quad t \geq 0.$$

It is $\mathcal{F}$ -measurable, since it is ρ -continuous. Also

$$B_t \circ \theta_s = B_{t+s}.$$

The process $((B_t)_{t \geq 0}, \Omega, \mathcal{F}, (\mathbb{P}_x)_{x \in \mathbb{R}})$ satisfies the Markov property with respect to $(\mathcal{F}_t^0)$:

$$\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^0] = \mathbb{E}_{B_s}[Y] \quad \mathbb{P}_x\text{-a.s.},$$

for all $s \geq 0$, $x \in \mathbb{R}$, and bounded $\mathcal{F}$ -measurable Y .

Let

$$\mathcal{F}_t^{0+} := \bigcap_{r>0} \mathcal{F}_{t+r}^0.$$

Then $(\mathcal{F}_t^{0+})$ is right-continuous, and the Markov property also holds with respect to $(\mathcal{F}_t^{0+})$:

$$\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^{0+}] = \mathbb{E}_{B_s}[Y] \quad \mathbb{P}_x\text{-a.s.}$$

Here $\mathbb{E}_{B_s}[Y]$ is $\mathcal{F}_s^0$ -measurable.

Proof. It is enough to prove the result for $Y = \mathbf{1}_A$, where A is a cylinder set

$$A = \{(B_{t_1}, \dots, B_{t_n}) \in A_1 \times \dots \times A_n\}, \quad A_i \in \mathcal{B}(\mathbb{R}).$$

By the monotone class theorem, this ends the proof.

Let $C \in \mathcal{F}_{s+h}^0$ be a cylinder set

$$C = \{(B_{s_1}, \dots, B_{s_m}) \in C_1 \times \dots \times C_m\}, \quad 0 \leq s_1 < \dots < s_m < s+h,$$

where $h < t_1$. Then

$$\begin{aligned} \mathbb{E}_x[\mathbf{1}_A \circ \theta_s \mathbf{1}_C] &= \mathbb{E}_x \left[\prod_{j=1}^m \mathbf{1}_{C_j}(B_{s_j}) \prod_{i=1}^n \mathbf{1}_{A_i}(B_{t_i+s}) \right] \\ &= \mathbb{P}_x((B_{s_1}, \dots, B_{s_m}, B_{s+t_1}, \dots, B_{s+t_n}) \in C_1 \times \dots \times C_m \times A_1 \times \dots \times A_n) \\ &= \mathbb{E}_x \left[\prod_{j=1}^m \mathbf{1}_{C_j}(B_{s_j}) F(B_{s+h}, h) \right] \\ &= \mathbb{E}_x[F(B_{s+h}, h) \mathbf{1}_C], \end{aligned}$$

where

$$F(y, h) = \int_{A_1} \dots \int_{A_n} \prod_{i=1}^n p_{t_i-t_{i-1}}(x_{i-1}, dx_i).$$

Thus

$$\mathbb{E}_x[\mathbf{1}_A \circ \theta_s \mathbf{1}_C] = \mathbb{E}_x[F(B_{s+h}, h) \mathbf{1}_C]$$

holds for cylinder sets $C \in \mathcal{F}_{s+h}^0$, and hence for all $C \in \mathcal{F}_{s+h}^0$ by the Dynkin system argument. Therefore it also holds for $C \in \mathcal{F}_s^{0+}$.

It remains to show

$$\lim_{h \downarrow 0} \mathbb{E}_x[F(B_{s+h}, h) \mathbf{1}_C] = \mathbb{E}_x[\mathbb{E}_{B_s}[\mathbf{1}_A] \mathbf{1}_C], \quad C \in \mathcal{F}_s^{0+}.$$

Here

$$\begin{aligned} F(y, h) &= \int_{A_1} \dots \int_{A_n} p_{t_1-h}(y, dx_1) \prod_{i=2}^n p_{t_i-t_{i-1}}(x_{i-1}, dx_i) \\ &= \int_{A_1} \dots \int_{A_n} \frac{1}{\sqrt{2\pi(t_1-h)}} \exp\left(-\frac{(y-x_1)^2}{2(t_1-h)}\right) \prod_{i=2}^n p_{t_i-t_{i-1}}(x_{i-1}, dx_i). \end{aligned}$$

This is jointly continuous in (y, h) and bounded by 1. As $h \downarrow 0$,

$$F(B_{s+h}, h) \rightarrow \int_{A_1} \dots \int_{A_n} p_{t_1}(B_s, dx_1) \prod_{i=2}^n p_{t_i-t_{i-1}}(x_{i-1}, dx_i) = \mathbb{E}_{B_s}[\mathbf{1}_A].$$

By the dominated convergence theorem, the claim holds. $\square$

Remark 2. The map $x \mapsto \mathbb{E}_x[Y]$ is $\mathcal{B}(\mathbb{R})$ -measurable. Therefore $\mathbb{E}_{B_s}[Y]$ is $\sigma(B_s)$ -measurable, and

$$\mathbb{E}_x[Y \circ \theta_s \mid \sigma(B_s)] = \mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^0].$$

Theorem 5. If $Z \in \mathcal{F}_b$, then

$$\mathbb{E}_x[Z \mid \mathcal{F}_s^{0+}] = \mathbb{E}_x[Z \mid \mathcal{F}_s^0] \quad \mathbb{P}_x\text{-a.s.}$$

for all $s \geq 0$ and $x \in \mathbb{R}$.

Proof. By a monotone class theorem, it is sufficient to prove this for $Z = \mathbf{1}_A$, where

$$A = \{(B_{t_1}, \dots, B_{t_n}) \in A_1 \times \dots \times A_n\}.$$

Let $\mathcal{M} = \{\mathbf{1}_A : \text{the equation holds}\}$. Then $\mathcal{M}$ is multiplicative. Using the Markov property,

$$\begin{aligned} \mathbb{E}_x[\mathbf{1}_A \mid \mathcal{F}_s^{0+}] &= \mathbb{E}_x \left[\prod_{t_i \leq s} \mathbf{1}_{A_i}(B_{t_i}) \prod_{t_i > s} \mathbf{1}_{A_i}(B_{t_i}) \mid \mathcal{F}_s^{0+} \right] \\ &= \prod_{t_i \leq s} \mathbf{1}_{A_i}(B_{t_i}) \mathbb{E}_x \left[\prod_{t_i > s} \mathbf{1}_{A_i}(B_{t_i-s}) \circ \theta_s \mid \mathcal{F}_s^{0+} \right] \\ &= \prod_{t_i \leq s} \mathbf{1}_{A_i}(B_{t_i}) \mathbb{E}_x \left[\prod_{t_i > s} \mathbf{1}_{A_i}(B_{t_i-s}) \circ \theta_s \mid \mathcal{F}_s^0 \right] \\ &= \mathbb{E}_x[\mathbf{1}_A \mid \mathcal{F}_s^0]. \end{aligned}$$

□

Theorem 6 (Blumenthal's 0-1 law). *If $A \in \mathcal{F}_0^{0+}$, then*

$$\mathbb{P}_x(A) = 0 \quad \text{or} \quad 1.$$

Proof. Since $\mathcal{F}_0^0 = \sigma(B_0)$, every $\mathcal{F}_0^0$ -measurable function is $\mathbb{P}_x$ -a.s. constant. Since $A \in \mathcal{F}_0^{0+}$,

$$\mathbf{1}_A = \mathbb{E}_x[\mathbf{1}_A \mid \mathcal{F}_0^{0+}] = \mathbb{E}_x[\mathbf{1}_A \mid \mathcal{F}_0^0] = \mathbb{P}_x(A).$$

Thus $\mathbf{1}_A$ is a constant equal to $\mathbb{P}_x(A)$, so $\mathbb{P}_x(A)$ is 0 or 1.

□

1.3 The d -Dimensional Brownian Motion

Define the transition density

$$p_t(x, y) = \frac{1}{(2\pi t)^{d/2}} \exp\left(-\frac{\|y - x\|^2}{2t}\right), \quad t > 0,$$

and

$$p_0(x, y)dy = \delta_x(dy), \quad t = 0.$$

Thus

$$\mu_{t_1, \dots, t_n}(A_1 \times \dots \times A_n) = \int_{A_1} \dots \int_{A_n} \prod_{i=1}^n p_{t_i - t_{i-1}}(x_{i-1}, dx_i).$$

By Kolmogorov's extension theorem,

$$\mu_{t_1, \dots, t_n} = \mathbb{P} \circ f_{t_1, \dots, t_n}^{-1}.$$

The d -dimensional Wiener measure is a copy of $\mathbb{P}_0$ on

$$(C(\mathbb{R}_+), \mathcal{B}(C(\mathbb{R}_+))).$$

Define

$$\mathbb{P}_0 = \bigotimes_{i=1}^d \mathbb{P}^{(i)}, \quad \Omega = C(\mathbb{R}_+)^d, \quad \mathcal{F} = \mathcal{B}(C(\mathbb{R}_+)^d).$$

Then

$$W_t(\omega) = \omega(t) = (\omega_1(t), \dots, \omega_d(t))$$

is canonical d -dimensional Brownian motion starting at 0. Also

$$\mathbb{P}_x = \mathbb{P}_0 \circ T_x^{-1}$$

is the law starting at x . Let

$$\mathcal{F}_t^0 = \sigma(W_s : s \leq t), \quad \mathcal{F}_t^{0+} = \bigcap_{r>0} \mathcal{F}_{t+r}^0.$$

Thus d -dimensional Brownian motion satisfies the Markov property:

$$\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^0] = \mathbb{E}_{W_s}[Y] \quad \mathbb{P}_x\text{-a.s.},$$

and also

$$\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^{0+}] = \mathbb{E}_{W_s}[Y] \quad \mathbb{P}_x\text{-a.s.}$$

If $Z \in \mathcal{F}_b$, then

$$\mathbb{E}_x[Z \mid \mathcal{F}_s^{0+}] = \mathbb{E}_x[Z \mid \mathcal{F}_s^0] \quad \mathbb{P}_x\text{-a.s.}$$

Blumenthal's 0–1 law also holds:

$$A \in \mathcal{F}_0^{0+} \implies \mathbb{P}_x(A) = 0 \text{ or } 1, \quad x \in \mathbb{R}^d.$$

Definition 2. A stochastic process (W_t) with state space $\mathbb{R}^d$ on some $(\Omega, \mathcal{F}, \mathbb{P}_x)$ is called *d-dimensional Brownian motion with start in x* if:

- (i) $\mathbb{P}_x(\{\omega \in \Omega : W_0(\omega) = x\}) = 1$;
- (ii) $W_t - W_s \sim N(0, |t - s|I)$;
- (iii) (W_t) has independent increments;
- (iv) almost all sample paths of (W_t) are continuous, i.e.

$$\mathbb{P}_x(t \mapsto W_t \text{ is continuous on } [0, \infty)) = 1.$$

Remark 3. A general d-dimensional Brownian motion satisfies the Markov property. For $A_i \in \mathcal{B}(\mathbb{R}^d)$,

$$\mathbb{E}_x \left[\prod_{i=1}^n \mathbf{1}_{A_i}(W_{t_i+s}) \mid \mathcal{F}_s^0 \right] = \mathbb{E}_{W_s} \left[\prod_{i=1}^n \mathbf{1}_{A_i}(W_{t_i}) \right] \quad \mathbb{P}_x\text{-a.s.}$$

Remark 4. Q -a.e. path in $\mathbb{R}^{\mathbb{Q}_+}$ is locally Hölder continuous of order $\gamma \in (0, 1/2)$ on $[0, n]$ for all n . Consequently, $\mathbb{P}$ -a.e. path of Brownian motion in $C(\mathbb{R}_+)$ is also locally Hölder continuous of order γ .

Remark 5. The process

$$((B_t)_{t \geq 0}, \Omega, \mathcal{F}, \mathbb{P})$$

defined above is called the canonical model of Brownian motion. The measure

$$\mathbb{P} = \mathbb{P}_0$$

is called Wiener measure. In this model,

- $B_t(\omega) = \omega(t)$;
- $\Omega = C(\mathbb{R}_+)$ and $\mathcal{F} = \sigma(B_t : t \geq 0)$;
- $\mathcal{F} = \mathcal{B}(C(\mathbb{R}_+))$;
- $\mathbb{P} = Q \circ \Psi^{-1}$.

2 The Strong Markov Property

2.1 Stopping Times and Filtrations

A filtered probability space is

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P}).$$

A filtration is *right-continuous* if

$$\mathcal{F}_t = \bigcap_{\varepsilon > 0} \mathcal{F}_{t+\varepsilon}.$$

We impose the usual conditions:

- (i) $(\mathcal{F}_t)_{t \geq 0}$ is right-continuous.
- (ii) $(\Omega, \mathcal{F}, \mathbb{P})$ is a complete probability space.
- (iii) If $N \in \mathcal{F}$ and $\mathbb{P}(N) = 0$, then $N \in \mathcal{F}_0$.

A stochastic process $X = (X_t)_{t \geq 0}$ is adapted to $(\mathcal{F}_t)_{t \geq 0}$ if X_t is $\mathcal{F}_t$ -measurable. The natural filtration of X is

$$\mathcal{F}_t^0 = \mathcal{F}_t^0(X) = \sigma(X_s : s \leq t).$$

The stochastic process is continuous if

$$\mathbb{P}(t \mapsto X_t \text{ is continuous on } [0, \infty)) = 1.$$

Example: canonical model of d -dimensional Brownian motion.

$$\mathbb{M}^W = (\Omega = C([0, \infty), \mathbb{R}^d), \mathcal{F} = \mathcal{B}(C([0, \infty), \mathbb{R}^d)), (W_t)_{t \geq 0}, (\mathbb{P}_x)_{x \in \mathbb{R}^d}),$$

with start at $x \in \mathbb{R}^d$. The process $(W_t)_{t \geq 0}$ has continuous sample paths, and

$$\mathcal{F}_t^0 = \mathcal{F}_t^0(W) = \sigma(W_s : s \leq t), \quad \mathcal{F}_t^{0+} = \mathcal{F}_{t+}^0(W) = \bigcap_{u>0} \sigma(W_s : s \leq t+u).$$

The completed right-continuous filtration is denoted by

$$\mathcal{F}_t^{0,x,+} := \text{completion of } \mathcal{F}_t^{0+} \text{ under } \mathbb{P}_x.$$

Definition 3 (Stopping time). A map $T : \Omega \rightarrow [0, \infty]$ is an $(\mathcal{F}_t)$ -stopping time if

$$\{T \leq t\} \in \mathcal{F}_t, \quad t \geq 0.$$

If $\{T < t\} \in \mathcal{F}_t$ for all $t \geq 0$, then we call T a weak stopping time.

If the filtration is right-continuous, then weak stopping time $\Rightarrow$ stopping time, since

$$\{T \leq t\} = \bigcap_{n \geq 1} \left\{ T < t + \frac{1}{n} \right\}.$$

Examples. If S, T are stopping times, then $S \vee T$, $S \wedge T$, and $S + T$ are stopping times.

If (T_n) is a sequence of stopping times, then $\sup_n T_n$ is a stopping time. Moreover, $\inf_n T_n$, $\limsup_n T_n$, and $\liminf_n T_n$ are weak stopping times.

Let $(X_t)_{t \geq 0}$ be $(\mathcal{F}_t)$ -adapted with values in a metric space (S, d) . Assume either

- (i) (X_t) has continuous sample paths and $(\mathcal{F}_t)$ is right-continuous; or
- (ii) (X_t) is continuous and $(\mathcal{F}_t)$ satisfies the usual conditions.

Measurability of an uncountable path is excluded by these assumptions.

For $A \subset S$, define

$$\sigma_A := \inf\{t > 0 : X_t \in A\}, \quad D_A := \inf\{t \geq 0 : X_t \in A\},$$

where $\inf \emptyset = \infty$. The time σ_A is the first hitting time, and D_A is the first entry time. Then

$$\{\sigma_A \geq t\} = \{X_s \notin A, \forall s \in (0, t)\}, \quad \{D_A \geq t\} = \{X_s \notin A, \forall s \in [0, t)\}.$$

If A is open, then $\sigma_A = D_A$ and these are weak stopping times. If A is closed, then D_A is a stopping time and σ_A is a weak stopping time.

For a closed set A ,

$$\begin{aligned} \{D_A \geq t\} &= \{X_s \notin A, \forall s \in [0, t]\} \\ &= \{d(X_s, A) > 0, \forall s \in [0, t]\} \\ &= \bigcup_{n \geq 1} \left\{ d(X_s, A) \geq \frac{1}{n}, \forall s \in [0, t] \right\} \\ &= \bigcup_{n \geq 1} \bigcap_{s \in [0, t] \cap \mathbb{Q}} \left\{ d(X_s, A) \geq \frac{1}{n} \right\} \in \mathcal{F}_t. \end{aligned}$$

For a stopping time T of $(\mathcal{F}_t)$, define

$$\mathcal{F}_T = \{A \in \mathcal{F} : A \cap \{T \leq t\} \in \mathcal{F}_t, \forall t \geq 0\}$$

This is a sub- σ -algebra.

- $\mathcal{F}_T$ is a σ -algebra and T is $\mathcal{F}_T$ -measurable.
- If S, T are stopping times and $S \leq T$, then $\mathcal{F}_S \subset \mathcal{F}_T$.
- If (T_n) is a decreasing sequence of stopping times with respect to a right-continuous filtration $(\mathcal{F}_t)$ and $T = \inf_n T_n$, then

$$\mathcal{F}_T = \bigcap_n \mathcal{F}_{T_n}.$$

Proof idea: $\mathcal{F}_T \subset \cap_n \mathcal{F}_{T_n}$ follows from $T \leq T_n$. Conversely, use

$$A \cap \{T < t\} = A \cap \bigcup_{n \geq 1} \{T_n < t\}.$$

Define

$$X_T(\omega) := X_{T(\omega)}(\omega).$$

Lemma 1. Let (X_t) be an $(\mathcal{F}_t)$ -adapted stochastic process with values in (S, d) . Assume either

- (i) (X_t) has continuous sample paths and $(\mathcal{F}_t)$ is right-continuous; or
- (ii) (X_t) is continuous and $(\mathcal{F}_t)$ satisfies the usual conditions.

If T is a finite stopping time, then X_T is $\mathcal{F}_T$ -measurable.

Proof idea: choose discretized stopping times $T_n \downarrow T$. Then $X_{T_n} \in \mathcal{F}_{T_n}$, and use continuity of paths.

Lemma 2. Let (X_t) be an $(\mathcal{F}_t)$ -adapted stochastic process with values in (S, d) , where $S \subset \mathbb{R}^d$ is Borel. Suppose that either (i) or (ii) above holds. Then

$$\mathbf{1}_{\{T < \infty\}} X_T$$

is $\mathcal{F}_T$ -measurable.

2.2 Markov Processes and the Strong Markov Property

Let S be a Borel subset of $\mathbb{R}^d$ containing 0.

Definition 4 (Transition kernels). A family

$$P_t : S \times \mathcal{B}(S) \rightarrow [0, 1]$$

is a Markovian semigroup of transition kernels on $(S, \mathcal{B}(S))$ if:

- (i) for all $t \geq 0$ and $x \in S$, $P_t(x, \cdot)$ is a probability measure;
- (ii) for all $t \geq 0$ and $A \in \mathcal{B}(S)$, the map $x \mapsto P_t(x, A)$ is $\mathcal{B}(S)$ -measurable;
- (iii) the Chapman-Kolmogorov equation holds:

$$P_t P_s = P_{t+s}, \quad P_t P_s(x, A) = \int_S P_s(y, A) P_t(x, dy) = P_{t+s}(x, A);$$

- (iv) $P_0(x, \cdot) = \delta_x$ for $x \in S$.

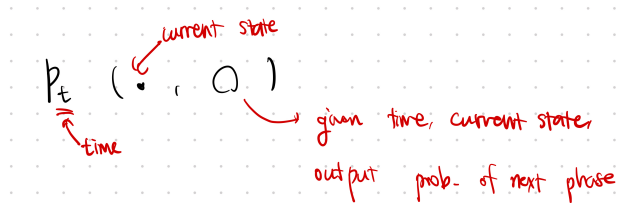


Figure 4: Interpretation of transition kernel

For bounded $\mathcal{B}(S)$ -measurable f , define

$$P_t f(x) = \int_S f(y) P_t(x, dy).$$

This is the expectation of f after time t , starting from x .

Definition 5 (Normal Markov process). A family

$$|M| = (\Omega, \mathcal{F}, (X_t)_{t \geq 0}, (\mathbb{P}_x)_{x \in S})$$

is a normal Markov process with respect to $(\mathcal{F}_t)$ on $(\Omega, \mathcal{F})$ if:

(M1) $x \mapsto \mathbb{P}_x(F)$ is $\mathcal{B}(S)$ -measurable for every $F \in \mathcal{F}$;

(M2) (X_t) is $(\mathcal{F}_t)$ -adapted and

$$\mathbb{P}_x(X_{s+t} \in A \mid \mathcal{F}_s) = \mathbb{P}_{X_s}(X_t \in A) \quad \mathbb{P}_x\text{-a.s.}$$

for all $s, t \geq 0$, $A \in \mathcal{B}(S)$, and $x \in S$;

(M3) $\mathbb{P}_x(X_0 = x) = 1$ for every $x \in S$.

We may change filtrations, so the filtration is not included as part of the intrinsic definition.

Theorem 7. Let $|M| = (\Omega, \mathcal{F}, (X_t)_{t \geq 0}, (\mathbb{P}_x)_{x \in S})$ be a family of stochastic processes, and let

$$\mathcal{F}_t^0 = \sigma(X_s : s \leq t).$$

(i) Suppose $(P_t)_{t \geq 0}$ is a Markovian semigroup of transition kernels such that, for $0 \leq t_1 < \dots < t_n$, $x \in S$, and $A_i \in \mathcal{B}(S)$,

$$\mathbb{P}_x((X_{t_1}, \dots, X_{t_n}) \in A_1 \times \dots \times A_n) = \int_{A_1} \dots \int_{A_n} \prod_{i=1}^n P_{t_i - t_{i-1}}(x_{i-1}, dx_i),$$

where $t_0 = 0$ and $x_0 = x$. Then $|M|$ is a normal Markov process with respect to the natural filtration $(\mathcal{F}_t^0)$.

(ii) Conversely, if $|M|$ is a normal Markov process and

$$P_t(x, A) := \mathbb{E}_x[\mathbf{1}_A(X_t)],$$

then (P_t) is a Markovian semigroup of transition kernels and the finite-dimensional distribution formula in (i) holds.

(iii) If (P_t) is a Markovian semigroup of transition kernels on $(S, \mathcal{B}(S))$, then there exists a normal Markov process $|M|$ satisfying

$$P_t(x, A) = \mathbb{E}_x[\mathbf{1}_A(X_t)]$$

with canonical path space $\Omega = S^{[0, \infty)}$.

The theorem gives the relation between a Markov process and its Markovian semigroup of kernels.

Proof. For (i), condition (M1) follows because $\mathcal{F} = \sigma(X_t : t \geq 0)$. It is enough to check events of the form $F = \{X_t \in A\}$:

$$\mathbb{P}_x(X_t \in A) = \int_A P_t(x, dx),$$

which is measurable in x . Also

$$\mathbb{P}_x(X_0 = x) = \int \delta_x(dx) = 1,$$

so (M3) holds.

For (M2), we must show

$$\mathbb{E}_x[\mathbf{1}_A(X_{s+t})\mathbf{1}_C] = \mathbb{E}_x[\mathbb{E}_{X_s}[\mathbf{1}_A(X_t)]\mathbf{1}_C], \quad C \in \mathcal{F}_s^0.$$

If $s = 0$, this is immediate. For $s > 0$, by the monotone class theorem it suffices to prove it for cylinder events

$$C = \{(X_{s_1}, \dots, X_{s_m}) \in C_1 \times \dots \times C_m\}, \quad 0 \leq s_1 < \dots < s_m = s.$$

Then

$$\begin{aligned} \mathbb{E}_x[\mathbf{1}_A(X_{s+t})\mathbf{1}_C] &= \mathbb{E}_x \left[\prod_{j=1}^m \mathbf{1}_{C_j}(X_{s_j}) \mathbf{1}_A(X_{s+t}) \right] \\ &= \int_{C_1} \dots \int_{C_m} \int_A \prod_{j=1}^{m+1} P_{s_j - s_{j-1}}(y_{j-1}, dy_j) \\ &= \int_{C_1} \dots \int_{C_m} \prod_{j=1}^m P_{s_j - s_{j-1}}(y_{j-1}, dy_j) \mathbb{E}_{y_m}[\mathbf{1}_A(X_t)] \\ &= \mathbb{E}_x \left[\mathbb{E}_{X_s}[\mathbf{1}_A(X_t)] \prod_{j=1}^m \mathbf{1}_{C_j}(X_{s_j}) \right] \\ &= \mathbb{E}_x[\mathbb{E}_{X_s}[\mathbf{1}_A(X_t)]\mathbf{1}_C]. \end{aligned}$$

For (ii), the identity

$$P_t(x, A) = \mathbb{E}_x[\mathbf{1}_A(X_t)]$$

clearly gives a probability measure in A , measurability in x , and $P_0(x, \cdot) = \delta_x$. For bounded f ,

$$P_t f(x) = \mathbb{E}_x[f(X_t)].$$

Moreover,

$$\begin{aligned} P_{t+s}(x, A) &= \mathbb{E}_x[\mathbf{1}_A(X_{t+s})] \\ &= \mathbb{E}_x[\mathbb{E}_x[\mathbf{1}_A(X_{t+s}) \mid \mathcal{F}_s]] \\ &= \mathbb{E}_x[\mathbb{E}_{X_s}[\mathbf{1}_A(X_t)]] \\ &= \mathbb{E}_x[P_t(X_s, A)] \\ &= \int_S P_t(y, A) P_s(x, dy). \end{aligned}$$

Thus the Chapman–Kolmogorov equation holds. □

Canonical model. Let $\Omega = S^{[0, \infty)}$. Think of elements as paths in S and write

$$X_t(\omega) = \omega(t), \quad \omega \in \Omega.$$

Let

$$\mathcal{F} = \sigma(X_t : t \geq 0), \quad \mathcal{F}_t^0 = \sigma(X_s : s \leq t).$$

The measures $\mathbb{P}_x$ are chosen so that (X_t) becomes a stochastic process satisfying

$$\mathbb{P}_x(X_0 = x) = 1.$$

The canonical shift is

$$\theta_s : \Omega \rightarrow \Omega, \quad (\theta_s \omega)(t) = \omega(s + t).$$

Let

$$M = (\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, (X_t)_{t \geq 0}, (\mathbb{P}_x)_{x \in S}, (\theta_t)_{t \geq 0})$$

be a canonical Markov process.

Lemma 3. Suppose $\mathbb{P}_x$ is given on the canonical model and satisfies the Markov property. Then (M2) holds for more general functions if and only if

$$\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s] = \mathbb{E}_{X_s}[Y] \quad \mathbb{P}_x\text{-a.s.},$$

for all $s \geq 0$, $x \in S$, and bounded $\mathcal{F}$ -measurable Y .

For a stopping time T , define

$$(\theta_T \omega)(t) = \omega(T(\omega) + t) \quad \text{if } T(\omega) < \infty,$$

and

$$(\theta_T \omega)(t) = \Delta \quad \text{if } T(\omega) = \infty, \quad f(\Delta) = 0.$$

Definition 6 (Strong Markov property). The process M has the strong Markov property if, for every $(\mathcal{F}_t)$ -stopping time T ,

$$\mathbb{E}_x[\mathbf{1}_{\{T < \infty\}} Y \circ \theta_T \mid \mathcal{F}_T] = \mathbf{1}_{\{T < \infty\}} \mathbb{E}_{X_T}[Y] \quad \mathbb{P}_x\text{-a.s.},$$

for all bounded $\mathcal{F}$ -measurable Y and $x \in S$.

If $T = s$ is constant, this is the usual formulation of the Markov property.

Theorem 8. Let

$$M = (\Omega, \mathcal{F}, (\mathcal{F}_t^{0+})_{t \geq 0}, (X_t)_{t \geq 0}, (\mathbb{P}_x)_{x \in S}, (\theta_t)_{t \geq 0})$$

be a canonical Markov process with right-continuous filtration. Assume:

(i) M has right-continuous sample paths;

(ii) the Feller property holds:

$$P_t(C_b(S)) \subset C_b(S).$$

Then M fulfills the strong Markov property with respect to $(\mathcal{F}_t)_{t \geq 0}$.

Proof. By a monotone class argument, it is enough to show the result for

$$Y = f_1(X_{t_1}) \cdots f_n(X_{t_n}), \quad f_i \in C_b(S).$$

Let $T_n \downarrow T$ be discrete stopping times and let $C \in \mathcal{F}_T$. By right-continuity,

$$\begin{aligned} \mathbb{E}_x[\mathbf{1}_{\{T < \infty\}} Y \circ \theta_T \mathbf{1}_C] &= \lim_{n \rightarrow \infty} \mathbb{E}_x[\mathbf{1}_{\{T < \infty\}} Y \circ \theta_{T_n} \mathbf{1}_C] \\ &= \lim_{n \rightarrow \infty} \sum_{k \geq 1} \mathbb{E}_x[\mathbf{1}_{C \cap \{(k-1)2^{-n} \leq T < k2^{-n}\}} Y \circ \theta_{k2^{-n}}]. \end{aligned}$$

By the Markov property,

$$\begin{aligned} &\lim_{n \rightarrow \infty} \sum_{k \geq 1} \mathbb{E}_x[\mathbf{1}_{C \cap \{(k-1)2^{-n} \leq T < k2^{-n}\}} \mathbb{E}_{X_{k2^{-n}}}[Y]] \\ &= \lim_{n \rightarrow \infty} \mathbb{E}_x[\mathbf{1}_{\{T < \infty\}} \mathbb{E}_{X_{T_n}}[Y] \mathbf{1}_C] \\ &= \mathbb{E}_x[\mathbf{1}_{\{T < \infty\}} \mathbb{E}_{X_T}[Y] \mathbf{1}_C]. \end{aligned}$$

Here $X_{T_n} \rightarrow X_T$, and $x \mapsto \mathbb{E}_x[Y]$ is continuous by the Feller property. □

More generally, let

$$Y : \mathbb{R}_+ \times \Omega \rightarrow S$$

be bounded and measurable, so that Y changes with time. Then

$$\mathbf{1}_{\{T < \infty\}} \mathbb{E}_{X_T}[Y(T, \cdot)] = \mathbf{1}_{\{T < \infty\}}(\omega) \mathbb{E}_{X_{T(\omega)}}[Y(T(\omega), \cdot)],$$

and

$$(Y(T, \cdot) \circ \theta_T)(\omega) = Y(T(\omega), \theta_T(\omega)).$$

Definition 7 (ESMP (Extended Strong Markov Property)). *Let*

$$M = (\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, (X_t)_{t \geq 0}, (\mathbb{P}_x)_{x \in S}, (\theta_t)_{t \geq 0})$$

be a canonical Markov process with right-continuous sample paths and right-continuous filtration $(\mathcal{F}_t)$. The process M has the extended strong Markov property (ESMP) if, for every $(\mathcal{F}_t)$ -stopping time T and every bounded $(\mathcal{B}(\mathbb{R}_+) \otimes \mathcal{F})$ -measurable Y ,

$$\mathbb{E}_x[\mathbf{1}_{\{T < \infty\}} Y(T, \cdot) \circ \theta_T \mid \mathcal{F}_T] = \mathbf{1}_{\{T < \infty\}} \mathbb{E}_{X_T}[Y(T, \cdot)] \quad \mathbb{P}_x\text{-a.s.}$$

Use the monotone class theorem for functions of the form $Y(t, \omega) = g(t)f(X_{t_1}(\omega))$.

Theorem 9. *If M has right-continuous sample paths and the Feller property*

$$P_t(C_b(S)) \subset C_b(S),$$

then M has the ESMP with respect to $(\mathcal{F}_t)$.

Remark 6. *The Brownian canonical process*

$$M^W = (\Omega, \mathcal{F}, (\mathcal{F}_t^{0+})_{t \geq 0}, (W_t)_{t \geq 0}, (\mathbb{P}_x)_{x \in S}, (\theta_t)_{t \geq 0})$$

has right-continuous paths and the Feller property. Hence M^W satisfies the SMP and the ESMP.

2.3 Application to Brownian Motion

2.3.1 The Reflection Principle

Let

$$M^B = (\Omega, \mathcal{F}, (\mathcal{F}_t^{0+})_{t \geq 0}, (B_t)_{t \geq 0}, \mathbb{P}_0, (\theta_t)_{t \geq 0}).$$

It satisfies the ESMP.

For $a > 0$,

$$\mathbb{P}_0(D_{\{a\}} < t, B_t > a) = \frac{1}{2} \mathbb{P}_0(D_{\{a\}} < t).$$

Here D_A denotes the first entry time

$$D_A = \inf\{t \geq 0 : X_t \in A\}.$$

Proof. Let

$$Y(s, \omega) = \mathbf{1}_{(-\infty, t)}(s) \mathbf{1}_{(a, \infty)}(B_{t-s}(\omega))$$

and

$$T = \inf\{0 \leq s < t : B_s = a\}.$$

Then

$$\{T < \infty\} = \{T < t\} = \{D_{\{a\}} < t\},$$

and

$$\mathbf{1}_{\{T < \infty\}} Y(T, \theta_T) = \mathbf{1}_{\{D_{\{a\}} < t\}} \mathbf{1}_{\{B_t > a\}}.$$

By ESMP,

$$\begin{aligned} \mathbb{P}_0(D_{\{a\}} < t, B_t > a) &= \mathbb{E}_0[\mathbf{1}_{\{T < \infty\}} Y(T, \theta_T)] \\ &= \mathbb{E}_0[\mathbf{1}_{\{T < t\}} \mathbb{E}_a[\mathbf{1}_{\{B_{t-T} > a\}}]] \\ &= \frac{1}{2} \mathbb{P}_0(T < t). \end{aligned}$$

Thus

$$\mathbb{P}_0(D_{\{a\}} < t) = 2\mathbb{P}_0(B_t > a).$$

Hence

$$\mathbb{P}_0(D_{\{a\}} < t) = \int_0^t \frac{|a|}{\sqrt{2\pi s^3}} \exp\left(-\frac{a^2}{2s}\right) ds,$$

and

$$\mathbb{P}_0\left(\sup_{s \leq t} B_s \leq a\right) = \mathbb{P}_0(|B_t| \leq a) = \frac{2}{\sqrt{2\pi t}} \int_0^a \exp\left(-\frac{x^2}{2t}\right) dx.$$

□

Arcsine law. Let

$$L = \sup\{t \leq 1 : B_t = 0\}.$$

Then

$$\begin{aligned} \mathbb{P}_0(L \leq s) &= \mathbb{E}_0[\mathbb{E}_0[\mathbf{1}_{\{B_{t-s} \neq 0, \forall t \in (s, 1)\}} \circ \theta_s \mid \mathcal{F}_s^{0+}]] \\ &= \mathbb{E}_0[\mathbb{E}_{B_s}[\mathbf{1}_{\{B_{t-s} \neq 0, \forall t \in (s, 1)\}}]] \\ &= \int_{\mathbb{R}} p_s(0, y) \mathbb{P}_y(\sigma_{\{0\}} > 1 - s) dy \\ &= \frac{2}{\pi} \arcsin(\sqrt{s}). \end{aligned}$$

2.3.2 Further Path Properties of Brownian Motion

Theorem 10 (Time inversion). *Let (B_t) be a standard Brownian motion and define*

$$X_t = \begin{cases} tB_{1/t}, & t > 0, \\ 0, & t = 0. \end{cases}$$

Then $(X_t)_{t \geq 0}$ is also a standard Brownian motion. Also, if (B_t) is a canonical Brownian motion, then so is M^X .

Corollary 1. *Let $\gamma \in (0, 1/2)$. For $\mathbb{P}_0$ -a.e. $\omega \in \Omega$, there exists $n^*(\omega) \in \mathbb{N}$ such that*

$$|X_t(\omega)| \leq Ct^\gamma, \quad t \leq 2^{-n^*(\omega)},$$

where C depends on γ . Hence, for $t \geq 2^{n^(\omega)}$,*

$$|B_t(\omega)| \leq C(1+t)^{1-\gamma}.$$

Theorem 11. *Let*

$$\mathcal{J} = \bigcap_{\varepsilon > 0} \mathcal{F}_\varepsilon^{0+}.$$

Then $\mathbb{P}_0$ is deterministic on $\mathcal{J}$, i.e.

$$\mathbb{P}_0(A) \in \{0, 1\}, \quad A \in \mathcal{J}.$$

Proof. Assume

$$\Omega = \left\{ \omega \in C(\mathbb{R}_+) : \lim_{t \rightarrow 0} t\omega\left(\frac{1}{t}\right) \in \mathbb{R} \right\}.$$

Then $X_t(\omega) = t\omega(1/t)$ converges as $t \rightarrow 0$. Moreover,

$$\sigma(X_s : 0 \leq s \leq t) = \sigma(B_s : s \geq 1/t).$$

Thus

$$\begin{aligned} \mathcal{J} &= \bigcap_{t>0} \sigma(B_s : s \geq t) \\ &= \bigcap_{t>0} \sigma\left(X_s : 0 \leq s \leq \frac{1}{t}\right) \\ &= \bigcap_{t>0} \sigma(X_s : 0 \leq s \leq t) = \mathcal{F}_0^{0+}(X). \end{aligned}$$

Blumenthal's 0–1 law gives the result. □

Theorem 12. *If (B_t) is canonical Brownian motion, then $\mathbb{P}_x$ -a.s.*

$$\limsup_{t \rightarrow \infty} \frac{B_t}{\sqrt{t}} = \infty, \quad \liminf_{t \rightarrow \infty} \frac{B_t}{\sqrt{t}} = -\infty.$$

Proof. For $K > 0$,

$$\left\{ \limsup_{n \rightarrow \infty} \frac{B_n}{\sqrt{n}} > K \right\} \in \mathcal{J},$$

and

$$\mathbb{P}_0\left(\frac{B_n}{\sqrt{n}} > K\right) = \mathbb{P}_0(B_1 > K) > 0.$$

The 0–1 law gives the claim. The lower-limit statement follows similarly. □